

STACK OVERFLOW DEVELOPER SURVEY 2019

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OUTLINE



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 - Visualization – Charts
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EXECUTIVE SUMMARY



- Built an end-to-end dashboard to identify this year's most in-demand developer skills.
- Top current languages: JavaScript, SQL, HTML/CSS, TypeScript, and Python.
- Databases are led by PostgreSQL; AWS dominates cloud platforms.
- Age 25–34 is the largest cohort; Bachelor's/Master's are the most common education levels.
- Deliverables: clean dataset, Cognos dashboards, and presentation-ready visuals.



INTRODUCTION



- Goal: Identify the most in-demand tech skills for this year to guide hiring, training, and roadmap decisions.
- Key Questions:
 - Which programming languages are most in demand?
 - Which database technologies are currently most sought after?
 - Which IDEs are the most popular?
- Scope: Programming languages, databases, web frameworks, and cloud platforms.
- Tools: Python + Jupyter for prep; IBM Cognos Analytics for dashboards/visuals.



METHODOLOGY

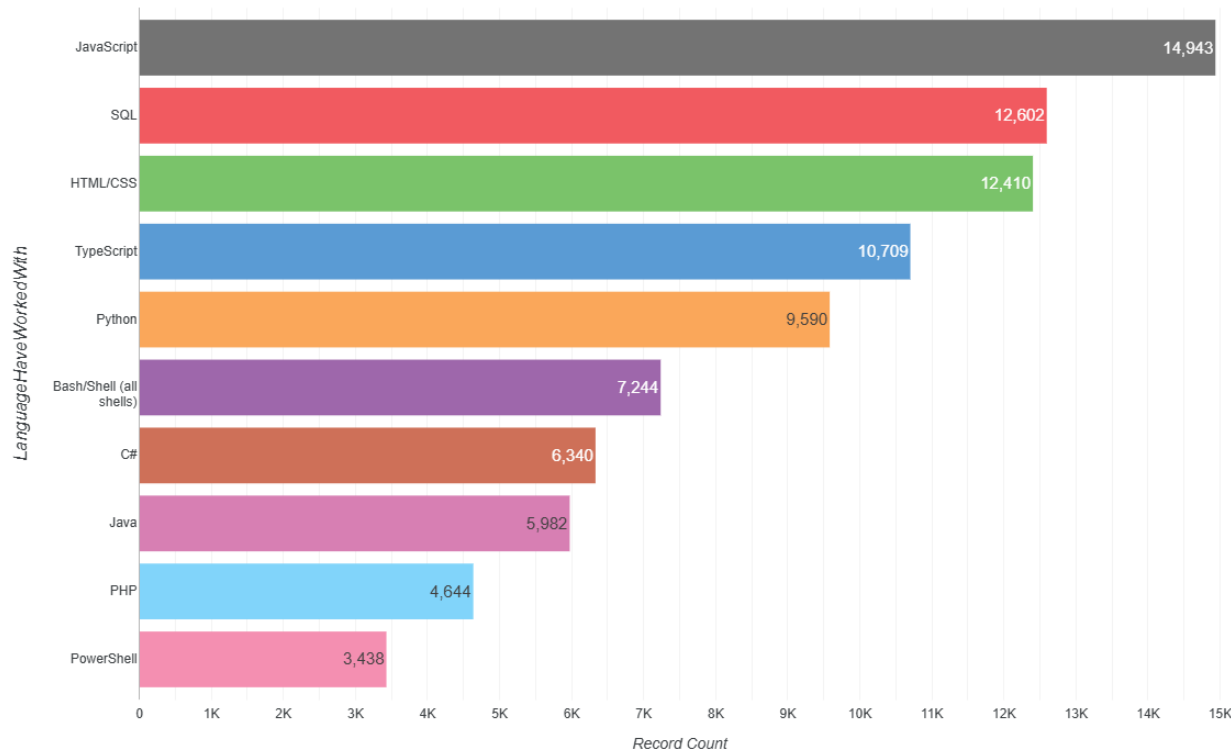


- Collect survey data & explore its content
 - Web Scraping
 - APIs.
 - Request library.
- Data Wrangling
- Exploratory data analysis
 - Analyzing data distribution.
 - Handling outliers.
 - Correlations.
- Data Visualization
 - Highlight distribution of data, relationships, the composition and comparison of data.
- Dashboards

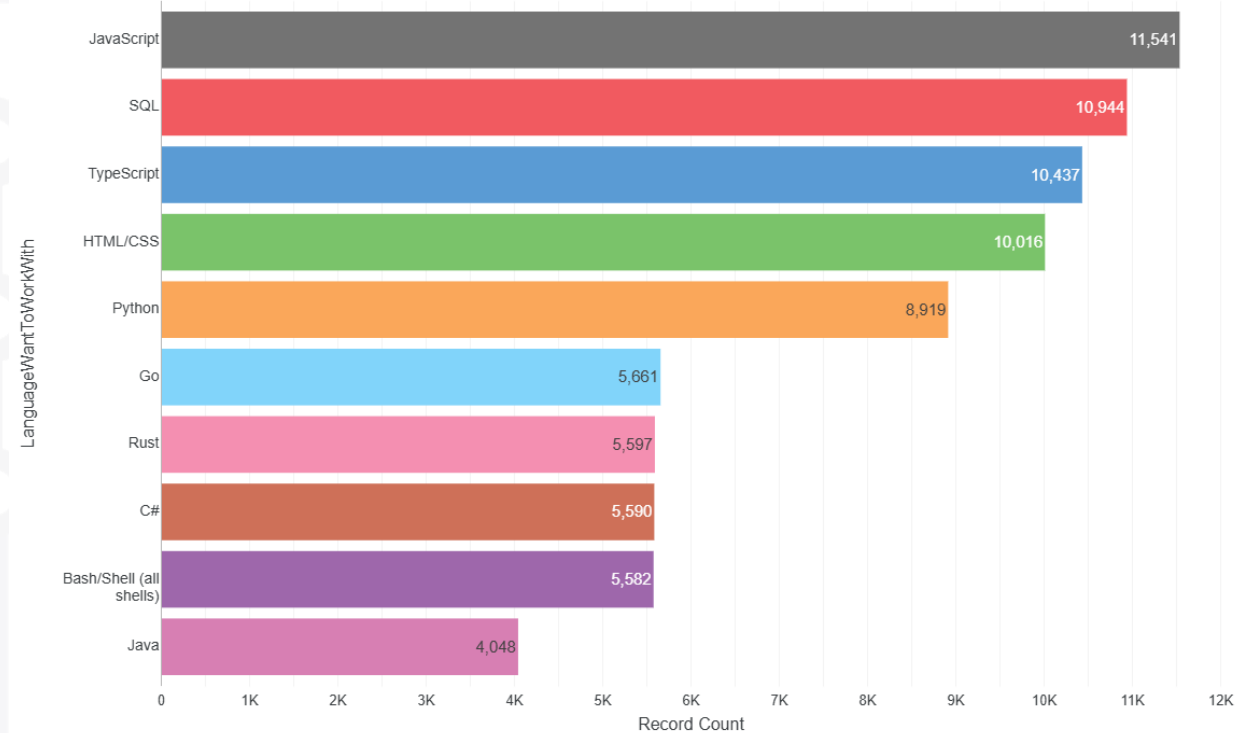


PROGRAMMING LANGUAGE TRENDS

Top 10 Languages Used



Top 10 Languages Desired Next Year



PROGRAMMING LANGUAGE TRENDS - FINDINGS & IMPLICATIONS

Findings

JavaScript, HTML/CSS & Python are the top 3 programming languages in 2025.

Python, JavaScript & SQL are the most popular programming languages respondents trend to anticipate next year.

JavaScript has remained the top trending language in the world for years

Implications

Python & JavaScript are critical for companies to invest.

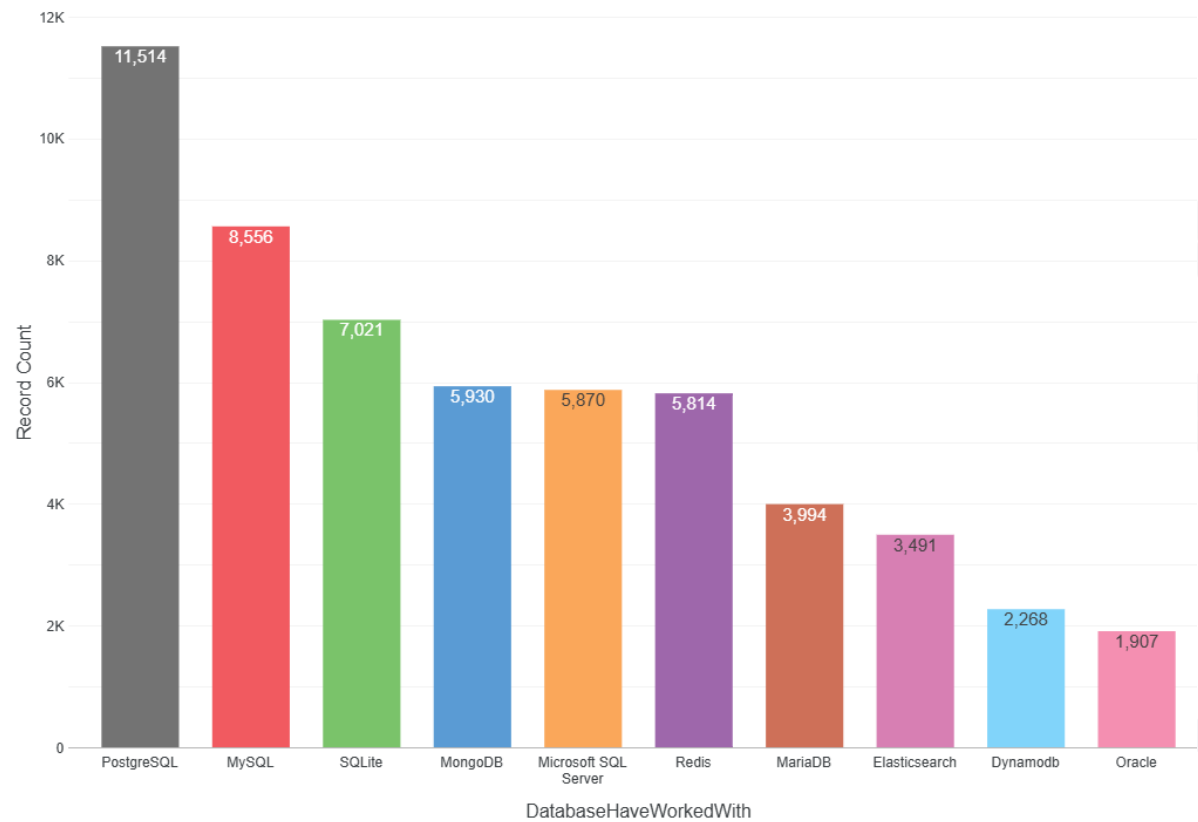
Consider SQL is the next language to invest to maintain the competitiveness in the market.

Web developments and Web developers are still in high demands

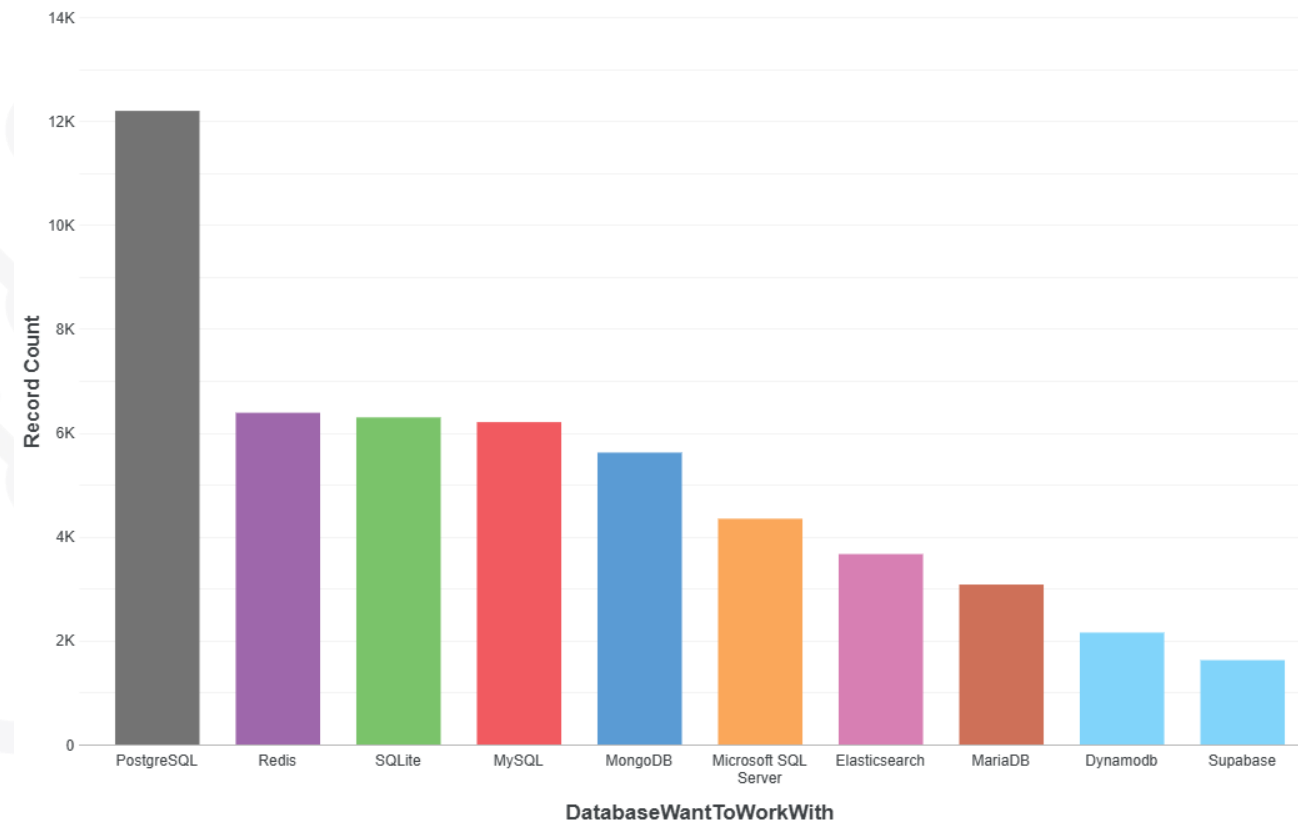


DATABASE TRENDS

Top 10 Databases Used



Top 10 Databases Desired Next Year



DATABASE TRENDS - FINDINGS & IMPLICATIONS

Findings

PostgreSQL, MySQL & SQLite are the top 3 databases in 2025.

The top 3 trends remain the same for the future.

Oracle is one of the top 10 database in 2025 but ruled out in the future. Instead, Supabase is raising.

Implications

Open source databases are popular among IT fields.

The demand of PostgreSQL is stable.

Relying on Oracle should be careful to start having alternative like Supabase for future investments.



DASHBOARD



Dashboard created using Google Looker

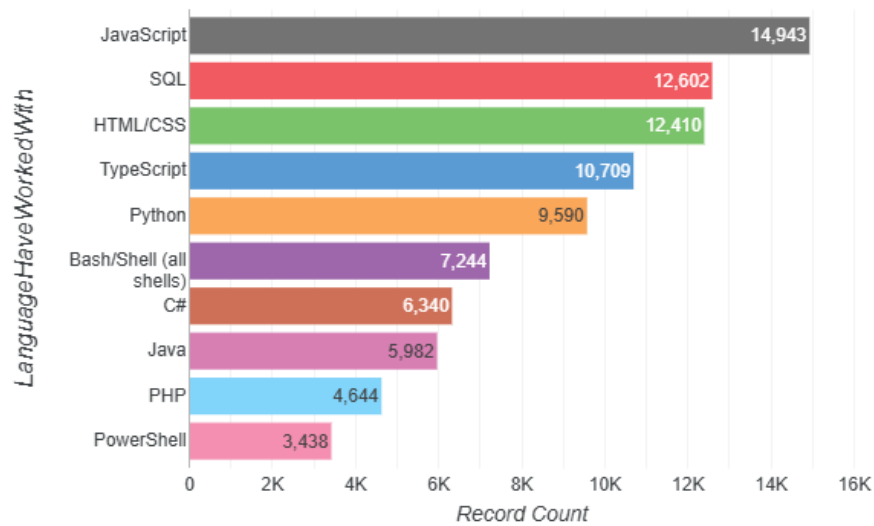
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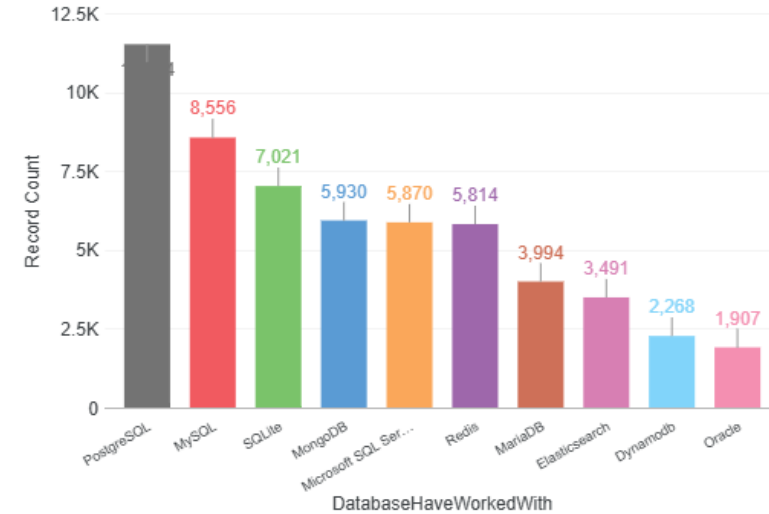
DASHBOARD TAB I

Current Technology Trends

Top 10 Languages Used



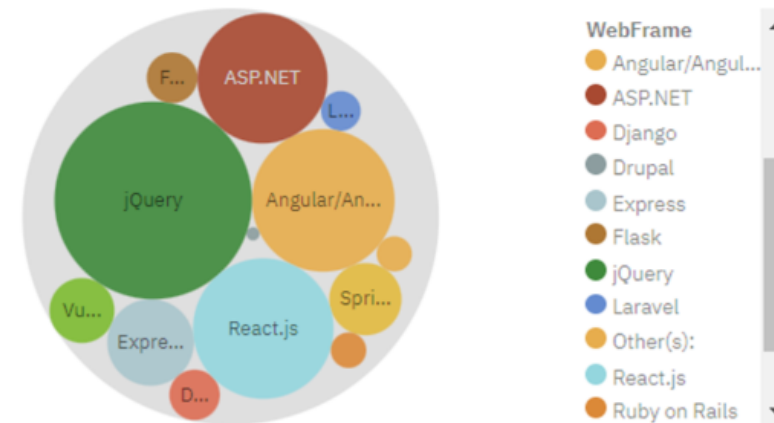
Top 10 Databases Used



Top 10 Platforms Used



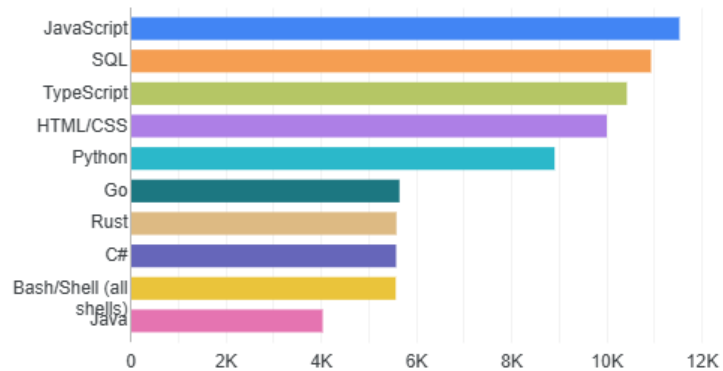
Top 10 WebFrames Used



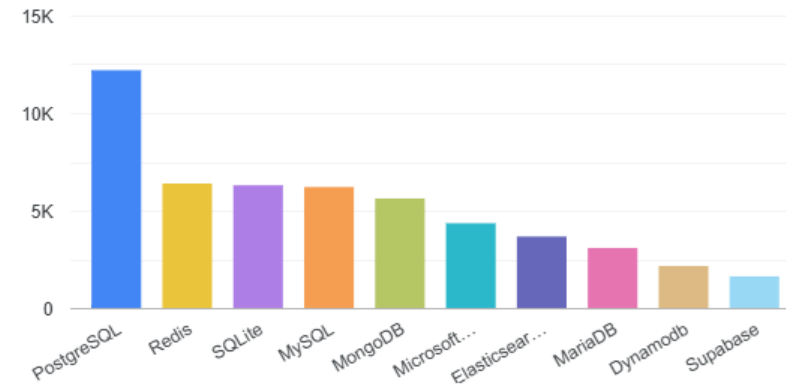
DASHBOARD TAB 2

Future Technology Trends

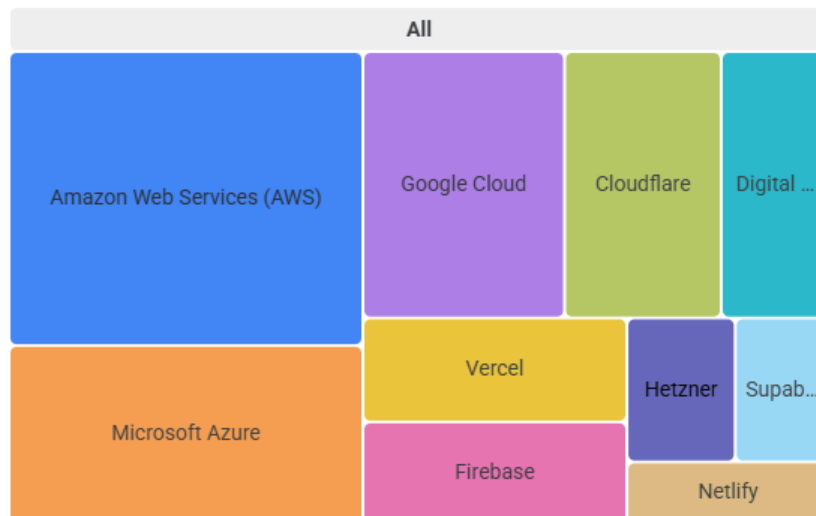
Top 10 Languages Desired Next Year



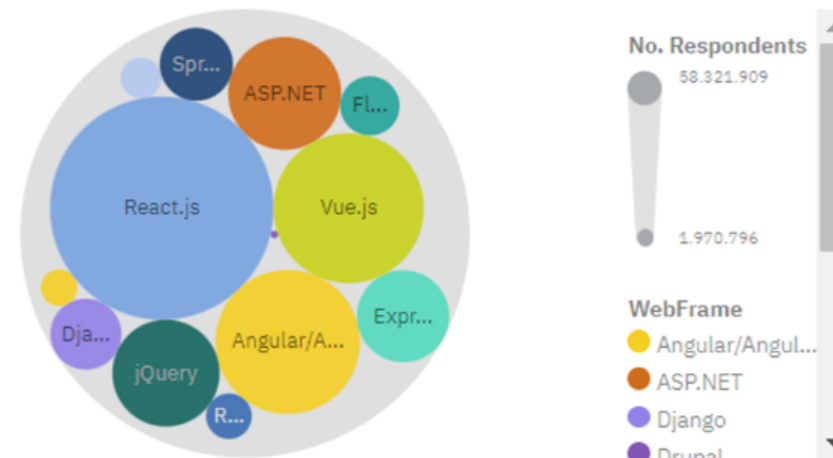
Top 10 Databases Desired Next Year



Top 10 Desired Platforms



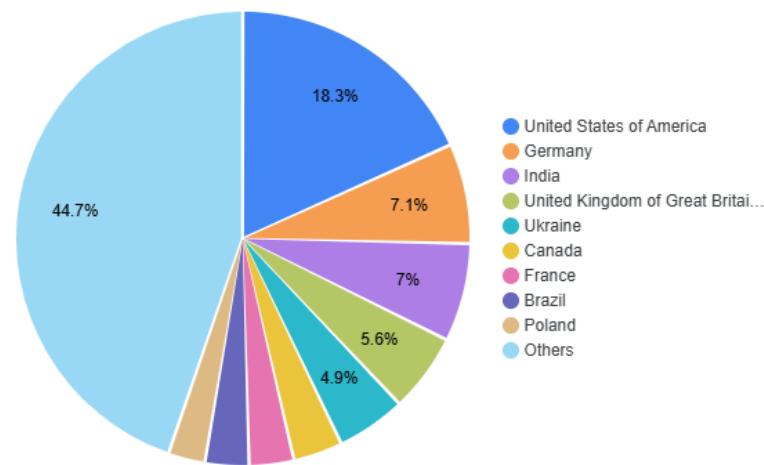
Top 10 WebFrame Next Year



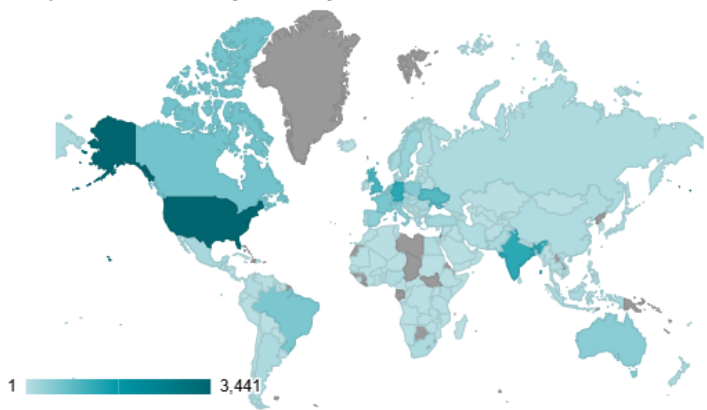
DASHBOARD TAB 3

Demographics

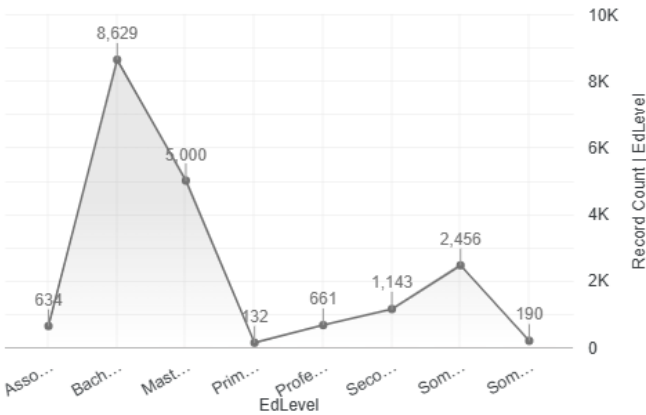
Respondents by Age



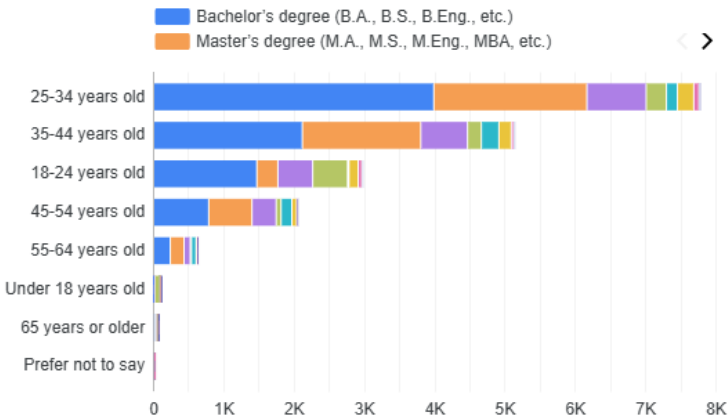
Respondent Count by Country



Respondent Distribution by Education Level



Respondent Count by Age, Classified by Education Level



Dashboard Insights

Current Trend

Language: HTML/CSS;JavaScript;TypeScript leads with 291 responses; C#;HTML/CSS;JavaScript;SQL;TypeScript follows with 240.

Database: PostgreSQL highest at 994.

Platform: AWS nearly 2,500 responses; Microsoft Azure ~2,000; Microsoft SQL Server at 689.

Webframe: Spring Boot highest at 398.

Future Trend

Language: React leads at 283; C#;HTML/CSS;JavaScript;SQL;TypeScript at 230; HTML/CSS;JavaScript;TypeScript at 217.

Database: PostgreSQL highest at 1,265.

Platform: AWS nearly 2,400; Microsoft SQL Server at 622.

Webframe: Spring Boot highest at 443.

Demographics

Platform: Microsoft Azure 1,649.

Age: 25–34 years highest (~8,000), followed by 35–44 years (~5,000).

Country: United States ~3,500 responses; Germany ~1,500.

Education: Bachelor's degrees (~8,500) highest, followed by Master's (~5,000).

Majority of respondents aged 25–34 hold Bachelor's or Master's degrees; similar but smaller patterns appear for 18–24, 35–44, and 45–54 groups.



OVERALL FINDINGS & IMPLICATIONS

Findings

Current: HTML/CSS + JavaScript + TypeScript dominates, C# + SQL combinations are gaining ground in the future.

PostgreSQL is the leader now and in the future, SQL Server is still #2 in the future but it is declining.

AWS is the most famous platform and Azure remains strong at 2nd

Age 25-34 is the dominant group with majority of Bachelor's degree holder, follow by Master's degree.

USA is the dominant countries in the IT fields follow by Germany

Implications

Recruitment and training efforts can be applied to age 25-34 group with Bachelor's degree for companies looking for IT positions and especially in USA and Germany. Also, companies should consider set up location(s) in USA and Germany

Companies invest on PostgreSQL should keep and put more resource while for SQL Server should maintain but look for alternative for backup

HTML/CSS + JavaScript + TypeScript combination is the strongest. Consider adding C# + SQL in the company to stay strong in the market.

Both AWS and Azure should be the long term platform for companies to invest

CONCLUSION



The IT industry is increasingly shifting toward an open-source, cloud-first ecosystem. PostgreSQL stands out as the most popular and in-demand database, both now and in the foreseeable future, while AWS and Microsoft Azure remain the dominant cloud platforms. Core programming languages such as Python, JavaScript, HTML/CSS, and SQL continue to define the landscape, with C# emerging as a growing contender worth strategic investment.

Organizations should align their long-term technology strategies with these open-source and cloud-first trends. From a talent perspective, the 25–34 age group holding Bachelor's or Master's degrees, primarily based in the United States and Germany, represents the largest segment of IT professionals. Recruitment and training efforts should be tailored to engage and develop this demographic for sustained growth.

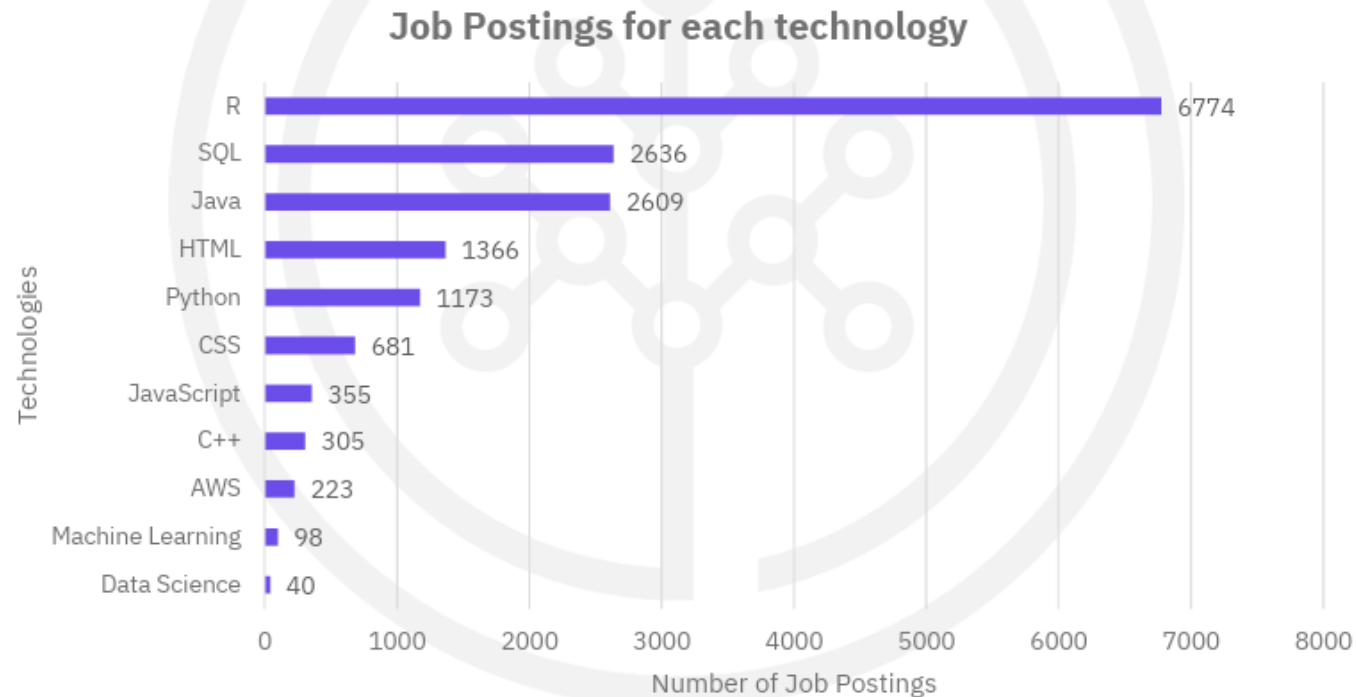


APPENDIX



JOB POSTINGS

In Module 1 you have collected the job posting data using Job API in a file named “job-postings.xlsx”. Present that data using a bar chart here. Order the bar chart in the descending order of the number of job postings.



POPULAR LANGUAGES

In Module 1 you have collected the job postings data using web scraping in a file named “popular-languages.csv”. Present that data using a bar chart here. Order the bar chart in the descending order of salary.

